to leave, they mistook a decorative door painted on a wall of the bar for the real thing. The patrons of the H-Bar, though, are well accustomed to people passing through walls and hardly take note of George and Gracie's abrupt departure.

A century ago, while Conrad and Freud were illuminating the heart and the soul of darkness, the German physicist Max Planck shed the first ray of light on quantum mechanics, a conceptual framework that proclaims, among other things, that the H-Bar experiences of George and Gracie—when scaled down to the microscopic realm—need not be attributed to clouded faculties. Such unfamiliar and bizarre happenings are typical of how our universe, on extremely small scales, actually behaves.

The Quantum Framework

Quantum mechanics is a conceptual framework for understanding the microscopic properties of the universe. And just as special relativity and general relativity require dramatic changes in our worldview when things are moving very quickly or when they are very massive, quantum mechanics reveals that the universe has equally if not more startling properties when examined on atomic and subatomic distance scales. In 1965, Richard Feynman, one of the greatest practitioners of quantum mechanics, wrote,

There was a time when the newspapers said that only twelve men understood the theory of relativity. I do not believe there ever was such a time. There might have been a time when only one man did because he was the only guy who caught on, before he wrote his paper. But after people read the paper a lot of people understood the theory of relativity in one way or other, certainly more than twelve. On the other hand I think I can safely say that nobody understands quantum mechanics.1

Although Feynman expressed this view more than three decades ago, it applies equally well today What he meant is that although the special and general theories of relativity require a drastic revision of previous ways of seeing the world, when one fully accepts the basic principles underlying them, the new and unfamiliar implications for space and time follow directly from careful logical reasoning. If you ponder the descriptions of Einstein's work in the preceding two chapters with adequate intensity, you will—if even for just a moment